

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/5791182>

# Muscle metabolic responses to exercise above and below the "critical power" assessed using $^{31}\text{P}$ -MRS

Article in *AJP Regulatory Integrative and Comparative Physiology* · February 2008

DOI: 10.1152/ajpregu.00731.2007 · Source: PubMed

CITATIONS

319

READS

648

5 authors, including:



**Andrew M Jones**

University of Exeter

398 PUBLICATIONS 19,963 CITATIONS

[SEE PROFILE](#)



**Daryl P Wilkerson**

University of Exeter

79 PUBLICATIONS 4,900 CITATIONS

[SEE PROFILE](#)



**Fred Dimenna**

Columbia University

54 PUBLICATIONS 3,571 CITATIONS

[SEE PROFILE](#)



**Jon Fulford**

University of Exeter

120 PUBLICATIONS 3,779 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Muscle vascular control mediators and their role in heart failure [View project](#)



Prolonged MV [View project](#)

# Muscle metabolic responses to exercise above and below the 'critical power' assessed using $^{31}\text{P}$ -MRS

Andrew M. Jones<sup>1</sup>, Daryl P. Wilkerson<sup>1</sup>, Fred DiMenna<sup>1</sup>, Jonathan Fulford<sup>2</sup>,  
and David C. Poole<sup>1,3</sup>

<sup>1</sup> School of Sport and Health Sciences and <sup>2</sup> Peninsula School of Medicine, St. Luke's Campus, University of Exeter, Heavitree Road, Exeter, Devon, EX1 2LU, UK; <sup>3</sup> Departments of Kinesiology, Anatomy and Physiology, Kansas State University, Manhattan, KS 66506, USA.

Correspondence:

Andrew M. Jones, Ph.D.

E-mail: [a.m.jones@exeter.ac.uk](mailto:a.m.jones@exeter.ac.uk)

Tel: 01392 262886

Fax: 01392 264726

Running Head:  $^{31}\text{P}$ -MRS assessment of critical power

## Abstract

We tested the hypothesis that the asymptote of the hyperbolic relationship between work-rate and the time-to-exhaustion during muscular exercise, the so-called ‘critical power’ (CP), represents the highest constant-work-rate that can be sustained without a progressive loss of homeostasis (as assessed using  $^{31}\text{P}$ -MRS measurements of muscle metabolites). Six healthy male subjects initially completed single-leg knee-extension exercise at 3-4 different constant-work-rates to the limit of tolerance (range: 3-18 min) for estimation of the CP (mean  $\pm$  SD,  $20 \pm 2$  W). Subsequently, the subjects exercised at work-rates that were 10% below CP (for 20 min) and 10% above CP (for as long as possible) while the metabolic responses in the contracting quadriceps muscle ([PCr],  $[\text{P}_i]$  and pH) were estimated using  $^{31}\text{P}$ -MRS. All subjects completed 20 min of exercise  $<$ CP without duress, whereas the limit of tolerance during  $>$ CP exercise was  $14.7 \pm 7.1$  min. During exercise  $<$ CP, stable values for [PCr],  $[\text{P}_i]$  and pH were attained within 3 minutes following the onset of exercise and there were no further significant changes in these variables (end-exercise [PCr]:  $68 \pm 11\%$  of baseline; end-exercise  $[\text{P}_i]$ :  $314 \pm 216\%$  of baseline; end-exercise pH:  $7.01 \pm 0.03$ ). During exercise  $>$ CP, however, [PCr] continued to fall to the point of exhaustion and  $[\text{P}_i]$  and pH changed precipitously to values that are typically observed at the termination of high-intensity exhaustive exercise (end-exercise [PCr]:  $27 \pm 17\%$  of baseline; end-exercise  $[\text{P}_i]$ :  $564 \pm 167\%$  of baseline; end-exercise pH:  $6.87 \pm 0.10$ ; all  $P < 0.05$  compared to  $<$ CP exercise). These data support the hypothesis that the CP represents the highest constant-work-rate that can be sustained without a progressive depletion of muscle high-energy phosphates and a rapid accumulation of metabolites (i.e.  $[\text{H}^+]$ ,  $[\text{P}_i]$ ) that have been associated with the fatigue process.

**Key Words:** fatigue threshold, maximal steady state, exercise tolerance,  $\text{O}_2$  kinetics

## Introduction

The hyperbolic relationship between work-rate and the time-to-exhaustion (defined here as an inability to maintain the external work-rate) is a fundamental property of exercise performance both in humans (37, 38, 44) and in other species (6, 15, 16, 34). Monod and Scherrer (37) first reported this hyperbolic relationship in a single muscle group and it was subsequently shown that this relationship was also demonstrable during whole-body exercise such as cycling (38, 63), treadmill running (24), swimming (60) and rowing (22). The work-rate asymptote of this hyperbolic relationship has been termed the ‘critical power’ (CP) or ‘fatigue threshold’; while the curvature constant, which represents the total amount of work that can be performed above the CP, has been termed the  $W'$  (37, 38, 44, 63). The parameters CP and  $W'$  can also be derived through linear regression analysis following transformation of the hyperbolic relationship into a linear formulation by plotting total work done during the series of exercise tests against time-to-exhaustion (37), or by plotting power output against the inverse of time-to-exhaustion (13, 44).

Moritani *et al.* (38) proposed that the CP represented the highest rate at which aerobic energy could be supplied during sustained exercise and that exercise below the CP could be sustained indefinitely (in reality, until the subject becomes limited by factors such as substrate depletion or hyperthermia). In contrast, sustained exercise above the CP requires utilization of the finite  $W'$  at a predictable rate until it is expended, at which point exercise cannot be continued at the same intensity. The  $W'$  therefore represents a constant amount of work that can be performed above the CP and is notionally equivalent to an energy store consisting of  $O_2$  reserves (myoglobin, venous blood), high-energy phosphates, and a source related to ‘anaerobic’ glycolysis (37, 38). The higher the sustained power output above the CP, the more rapidly the  $W'$  will be expended, and the greater will be the rate at which metabolites which have been associated with the fatigue process accumulate (e.g.,  $P_i$ , ADP,  $H^+$ , extra-cellular  $K^+$ ; 11, 48, 61, 66).

In support of the theoretical constructs of CP and  $W'$ , studies have shown that the CP is increased by endurance training (25, 45) and reduced by the inspiration of a hypoxic gas (38); and that the  $W'$  is increased by creatine supplementation (35, 55) and high-intensity strength and sprint training (26), but reduced following glycogen depletion (36) and prior high-intensity exercise with limited recovery (19). Moreover, the CP appears to broadly correspond to the exercise intensity at the

maximal lactate steady state (i.e., the highest power output that can be sustained without a continued increase in blood [lactate] with time; 47, 54). Indeed, Poole *et al.* (44) demonstrated that arterialized venous blood [lactate], [bicarbonate], and pH ultimately attained steady values when subjects exercised for 24 min at the CP, but that blood [lactate] increased and pH and [bicarbonate] fell progressively during exercise performed at a work-rate that was 5% above CP such that exhaustion occurred before the 24 min target was attained. Moreover, a steady-state in  $\dot{V}_{O_2}$  was attained at CP (albeit delayed and elevated by the  $\dot{V}_{O_2}$  slow component above the expected value), whereas during exercise above the CP,  $\dot{V}_{O_2}$  increased inexorably throughout exercise until a  $\dot{V}_{O_2}$  that was not different from the independently-determined  $\dot{V}_{O_2}$  max was reached (44).

On the basis of these distinct blood acid-base and pulmonary gas exchange profiles, the CP has been accepted as an important demarcator of exercise intensity with corresponding implications for the predominant mechanism(s) of muscular fatigue and the tolerable duration of exercise (28, 64). However, to our knowledge, no previous studies have investigated the temporal profile of the muscle metabolic responses to prolonged exercise performed <CP and > CP. If the CP is a valid theoretical and practical construct, one would predict that exercise <CP would result, after an initial rapid change, in a stabilization of phosphorylcreatine and inorganic phosphate concentrations ([PCr] and [P<sub>i</sub>], respectively) and pH in the contracting muscles. During exercise >CP where the W' will be utilized at a predictable rate until fatigue prevents the continuation of exercise, however, a continued fall in muscle [PCr] and pH and a continued rise in muscle [P<sub>i</sub>] would be expected. The concentration of muscle high-energy phosphate metabolites and pH can be assessed non-invasively, in real-time, and with high temporal resolution using <sup>31</sup>P phosphate magnetic resonance spectroscopy (<sup>31</sup>P-MRS).

In the present study, we therefore sought to investigate the mechanistic bases for the CP concept by using <sup>31</sup>P-MRS to resolve the muscle metabolic responses to constant-work-rate single-leg knee-extension exercise performed approximately 10% below and 10% above the pre-determined CP. We hypothesized that exercise <CP would result, after an initial rapid change, in the early attainment of stable values of [PCr], [P<sub>i</sub>], and pH, and the completion of 20 min of exercise without appreciable duress. We further hypothesized that exercise >CP would be characterized by a continued increase in [P<sub>i</sub>] and a continued reduction in [PCr] and pH until the subjects reached their limit of tolerance.

## Methods

### *Subjects*

Six male subjects (mean  $\pm$  SD age  $30 \pm 8$  years, stature  $1.82 \pm 0.04$  m, mass  $82.5 \pm 7.6$  kg) volunteered and gave written informed consent to participate in this study which had been approved by the local research ethics committee. The subjects were all recreationally-active in a variety of sport and exercise activities, were non-smokers, and were familiar with the exercise mode and experimental procedures used in the present study. The subjects were instructed to report to the laboratory in a rested state, having completed no strenuous exercise within the previous 24 hours, and having consumed a light carbohydrate-based meal 2-3 hours before the scheduled appointment. The subjects were also instructed to abstain from alcohol and caffeine for the preceding 12 hours and 6 hours, respectively. Adherence to these instructions was verified upon subjects' arrival at the laboratory.

### *Experimental Procedures*

The study was conducted in two parts. The work-rate vs. time-to-exhaustion relationship was first established during single-leg knee-extension exercise for each subject from 3-4 separate exercise bouts. From this relationship, the CP (and  $W'$ ) was estimated. Secondly, subjects performed single-leg knee-extension exercise at work-rates that were approximately 10% below and 10% above their CP while positioned inside the bore of a 1.5 T superconducting magnetic resonance scanner for  $^{31}\text{P}$ -MRS assessment of muscle metabolic responses. In all cases, the right leg was used for the experiments and subjects exercised using the same ergometer in both parts of the study. All exercise tests were conducted on separate days and at a similar time of day (within  $\pm 3$  hours).

#### *I: Derivation of the work-rate vs. time-to-exhaustion relationship and estimation of CP*

The subjects initially completed a minimum of three exercise bouts, each at a constant but different work-rate for determination of the hyperbolic relationship between work-rate and the time-to-exhaustion. The work-rates were selected, on the basis of previous tests performed by these same subjects during other experiments, to result in exhaustion in 2-15 min. Specifically, the highest and

lowest work-rates in the series were selected to elicit exhaustion in 2-3 min and 12-15 min, respectively, with the other work-rate(s) selected to cause exhaustion in ~ 4-8 min. In four subjects, the first three work-rates chosen produced times-to-exhaustion within the desired range, but in the remaining two subjects an additional work-rate was required to ensure an appropriate range of times-to-exhaustion. The exercise bouts were performed on separate days and were presented in random order. During all exercise tests, the subjects were verbally encouraged to continue exercising for as long as possible. The time-to-exhaustion, defined as the time at which the subject could no longer keep pace with the required rate of muscle contraction (see below), was recorded to the nearest second using a hand-held stopwatch.

The CP and  $W'$  were estimated using three separate models, namely: 1) the non-linear power-time model ( $\text{time} = W'/(P - CP)$ ), in which the asymptote of the hyperbolic relationship is the CP and the curvature constant is the  $W'$ ; 2) the linear work-time model in which work done is plotted against the time-to-exhaustion ( $W = CP \cdot \text{time} + W'$ ) and linear regression analysis is used to estimate the CP (the slope of the line) and the  $W'$  (the y intercept); and 3) the linear 1/time model in which work-rate is plotted against the inverse of the time-to-exhaustion and linear regression analysis is used to estimate the CP (the y intercept) and the  $W'$  (the slope of the line), (13, 17, 23, 40, 44, 47). The 95% confidence intervals for the estimation of CP from model 3 were used to calculate work-rates that were just below (~ 10%) and just above (~ 10%) the CP. For example, if the CP was estimated as 25 W with a 95% confidence interval of  $\pm 2$  W, then the subject exercised at a work-rate of 22 W during the <CP condition and at a work-rate of 28 W during the >CP condition. We reasoned that this approach would provide reasonable surety that the work-rates selected were actually just below and above the CP for all subjects.

## *II: $^{31}\text{P}$ -MRS assessment of muscle metabolic responses to <CP and >CP exercise*

Within 7 days of the completion of the predictive trials for estimation of the CP, the subjects reported to the MRS laboratory on two separate occasions to perform <CP and >CP exercise with simultaneous measurement of muscle metabolic responses using  $^{31}\text{P}$ -MRS. The order in which the <CP and >CP trials was presented was randomized. Subjects were required to complete 20 min of exercise <CP and to exercise for as long as possible >CP. With the exception of the measurement of the muscle metabolic response, the equipment and the general procedures were identical to those

used in the first part of the study. The time-to-exhaustion for exercise >CP was predicted (i.e., time =  $W'/(P-CP)$ ) and compared to the actual time-to-exhaustion.

### *Equipment and MRS measurements*

The single-leg knee-extension exercise tests were conducted in the prone position with subjects secured to the ergometer bed with Velcro straps at the thigh, buttocks and lower back to minimise extraneous movement during the protocol. The ergometer was constructed in-house and consists of a self-supporting nylon frame which fits onto the the bed, and is placed close to the subjects feet, and a base unit placed at the end of the bed. The subject's right foot was connected to a rope which runs over the top of the frame to the base unit on which a mounted pulley system permitted non-magnetic weights to be lifted and lowered. Within the pulley system a shaft encoder (Type BDK-06, Baumer Electronics, Swindon, UK) was fitted to record distances the load was moved through, alongside a non-magnetic load cell (Type F250, Novatech Measurements Ltd, St Leonards-On-Sea, UK) to record applied forces. Exercise was performed at a rate of 40 repetitions per minute with the subjects lifting and lowering the weight over a distance of  $\sim 0.22$  m in accordance with a visual cue. The equipment and procedures were the same in both parts of the study except that, in the second part, subjects were positioned inside a whole body MRI system. A 6 cm  $^{31}\text{P}$  transmit /receive surface coil was placed within the subject bed and the subject asked to lie upon it such that the coil was centred over the quadriceps muscle of the leg to be exercised.

MRS was performed in the Peninsula Magnetic Resonance Research Centre, Exeter, using a 1.5 T superconducting MR scanner (Intera, Philips, Netherlands). Initially, fast field echo images were acquired to determine that the muscle was positioned correctly relative to the coil. This was aided by placing cod liver oil capsules, which yield high intensity signal points within the image, adjacent to the coil, allowing its orientation relative to the muscle volume under examination to be assessed. A number of pre-acquisition steps were carried out in order to optimise the signal from the muscle under investigation. An automatic shimming protocol was undertaken within a volume that defined the quadriceps muscle in order to optimise the homogeneity of the local magnetic field, thereby leading to maximum signal collection. Tuning and matching of the coil was then performed to



maximise energy transfer between the coil and the muscle. To ensure that the examined muscle was consistently scanned with the same level of contraction, thereby ensuring a consistent distance from the coil at the time of data sampling, the subject was visually cued via a display consisting of 2 vertical bars, one which moved at a constant rate with a frequency of 0.67 Hz and one which monitored foot movement via a sensor present within the pulley to which they were connected. Thus, the subject endeavoured to match the movements of these two bars. The contraction phase of the knee extensors and  $^{31}\text{P}$ -MRS interrogation of the quadriceps was synchronised. At the time of data acquisition a trigger pulse was sent from the phosphorous MR amplifier to the visual display ensuring the two maintain synchronization. Thus, provided that subjects correctly match their movements to the display, the leg will always be in the same position when the signal is collected. The work done by the subject was recorded via a non-magnetic strain gauge present within the pulley mechanism, allowing the actual work-rate to be calculated.

Prior to exercise, during exercise and during recovery, data were acquired every 1.5 seconds, with a spectral width of 1500 Hz, and 1K data points. Phase cycling with 4 phase cycles was employed, leading to a spectra being acquired every 6 s. The subsequent spectra were quantified via peak fitting, assuming prior knowledge, using the jMRUI (version 2) software package and the AMARES fitting algorithm (58). Spectra were fitted assuming the presence of the following peaks:  $\text{P}_i$ , PCr,  $\alpha$ -ATP (2 peaks, amplitude ratio 1:1),  $\gamma$ -ATP (2 peaks, amplitude ratio 1:1),  $\beta$ -ATP (3 peaks, amplitude ratio 1:2:1), and phosphodiester. In all cases, relative amplitudes were corrected for partial saturation due to the repetition time relative to  $T_1$ . Intracellular pH was calculated using the chemical shift of the  $\text{P}_i$  spectral peak relative to the PCr peak (56).

The [PCr] and [ $\text{P}_i$ ] data were expressed as the % change relative to the resting baseline which was assumed to represent 100%. The change in [PCr] from rest to end-exercise for both the <CP and >CP conditions was divided by the respective work-rate to provide information on the response 'gain', that is, the change ( $\Delta$ ) in [PCr] per unit increase in work-rate. Also, to investigate the existence of a [PCr] 'slow component' during exercise >CP, the difference in [PCr] between 3 minutes and 6 minutes of exercise was computed.

## Statistics

<sup>31</sup>P-MRS data ([PCr], [P<sub>i</sub>], and pH) for the <CP and >CP conditions were expressed both at specific absolute time points (baseline, 1 min, 3 min, 6 min, 10 min and end-exercise), and as a fraction (0%, 25%, 50%, 75%, and 100%) of the time-to-exhaustion (for >CP) or time-to-end-exercise (20 min; for <CP). Statistical analysis of measurements was performed using ANOVA. Two two-way ANOVAs were used, one with repeated measures across intensity (<CP and >CP) and absolute time (0 min, 1 min, 3 min, 6 min, 10 min and end-exercise), and one with repeated measures across intensity (<CP and >CP) and relative time (0%, 25%, 50%, 75%, and 100%). When intensity-by-time interactions were significant, post-hoc one-way ANOVAs were undertaken on the relevant data and Bonferroni-adjusted paired t-tests were used as appropriate to identify differences between responses at specific time points. A paired t-test and a Pearson product moment correlation coefficient were used to investigate the relationship between the predicted and actual times-to-exhaustion during exercise >CP. A paired t-test was used to assess differences in the change in [PCr] between 3 and 6 minutes of exercise performed below and above the CP. Significance was accepted when  $P < 0.05$ . Data are expressed as mean  $\pm$  SD unless stated otherwise.

## Results

### *I: Derivation of the work-rate vs. time-to-exhaustion relationship and estimation of CP*

All subjects successfully completed a minimum of three constant-work-rate exercise trials for the estimation of CP. There was no significant difference in the parameter estimates derived from the three models either for CP (Model 1:  $19.6 \pm 2.2$  W; Model 2:  $19.8 \pm 1.9$  W; Model 3:  $19.9 \pm 1.8$  W;  $F_{2,5} = 1.75$ ;  $P = 0.22$ ) or  $W'$  (Model 1:  $2.09 \pm 0.76$  kJ; Model 2:  $2.02 \pm 0.65$  kJ; Model 3:  $1.98 \pm 0.60$  kJ;  $F_{2,5} = 1.27$ ;  $P = 0.32$ ). The work done vs. time and work-rate vs. 1/time relationships were highly linear for all subjects ( $R^2 = 0.99 \pm 0.01$ ). The hyperbolic relationship between work-rate and time-to-fatigue, along with the linear transformation of the relationship of work-rate to 1/time-to-exhaustion, is shown for a representative subject in Figure 1. The subjects' individual CP

and  $W'$  values (derived from Model 3), along with the work-rates applied in the second part of the study, are shown in Table 1.

## *II: $^{31}\text{P}$ -MRS assessment of muscle metabolic responses to <CP and >CP exercise*

During <CP exercise, all subjects were able to complete the prescribed 20 min exercise period without difficulty. However, during >CP exercise, the time-to-exhaustion was  $14.7 \pm 7.1$  min. This value was not significantly different from the predicted time-to-exhaustion for the work-rate ( $15.1 \pm 3.3$  min) and the actual and predicted times-to-exhaustion were significantly correlated ( $r = 0.87$ ;  $P < 0.05$ ).

The muscle metabolic responses for <CP and >CP exercise are summarised in Tables 2 and 3 and illustrated in Figures 2-4 (see below).

[PCr]. The two-way ANOVA with repeated measures across intensity and absolute time revealed a significant intensity-by-time interaction ( $F_{5,25} = 31.29$ ;  $P < 0.01$ ) and significant main effects for both intensity ( $F_{1,5} = 42.26$ ;  $P < 0.01$ ) and time ( $F_{5,25} = 44.07$ ;  $P < 0.01$ ). The subsequent one-way ANOVAs with repeated measures across time also indicated the existence of significant differences (<CP,  $F_{5,25} = 25.5$ ;  $P < 0.01$ ; >CP,  $F_{5,25} = 47.37$ ;  $P < 0.01$ ). The ANOVA results were essentially the same when data were expressed as a fraction of the time at the end of exercise, and are thus not reported. The pertinent statistical comparisons are presented in Table 2 (for absolute time) and Table 3 (for relative time). For the <CP exercise bout, following an initial fall, muscle [PCr] stabilised at  $\sim 75\%$  of the baseline value after  $\sim 3$  min. There was no significant change in muscle [PCr] between 3 min and the end of exercise at this intensity. In contrast, during >CP exercise, muscle [PCr] did not stabilise but rather continued to fall with time until exercise was terminated (Table 2). The muscle [PCr] at exhaustion at this intensity was  $26 \pm 16\%$  of the baseline value. Muscle [PCr] was significantly lower during >CP compared to <CP exercise at 1, 3, 6 and 10 min, and at the end of exercise ( $P < 0.05$ ). The change in [PCr] between rest and end-exercise per unit work-rate was almost twice as great for >CP exercise (<CP:  $1.86 \pm 0.55$  vs. >CP:  $3.40 \pm 0.82$   $\% \Delta[\text{PCr}] \cdot \text{W}^{-1}$ ;  $P < 0.01$ ). The difference in [PCr] between 3 and 6 minutes of exercise (used here as an index of a possible 'slow component' in the response) was significantly different between the two intensities (<CP:  $4 \pm 3\%$  vs. >CP:  $14 \pm 8\%$ ;  $P < 0.05$ ). The muscle [PCr] responses for <CP and

>CP exercise are shown in Figure 2 for a representative subject (Panel A) and for the entire group when the data are plotted as a function of the time-to-exhaustion (Panel B).

[ $P_i$ ]. The two-way ANOVA with repeated measures across intensity and absolute time revealed a significant intensity-by-time interaction ( $F_{5,25} = 4.99$ ;  $P < 0.01$ ) and significant main effects for both intensity ( $F_{1,5} = 62.58$ ;  $P < 0.01$ ) and time ( $F_{5,25} = 19.75$ ;  $P < 0.01$ ). The subsequent one-way ANOVAs with repeated measures across time also indicated the existence of significant differences (<CP,  $F_{5,25} = 6.91$ ;  $P < 0.01$ ; >CP,  $F_{5,25} = 21.34$ ;  $P < 0.01$ ). The ANOVA results were essentially the same when data were expressed as a fraction of the time at the end of exercise. The pertinent statistical comparisons are therefore presented in Table 2 (for absolute time) and Table 3 (for relative time). For the <CP exercise bout, muscle [ $P_i$ ] rose immediately following the onset of exercise but then reached a stable level after  $\sim 1$  min. There was no significant further increase in [ $P_i$ ] between 1 min ( $270 \pm 56\%$  of baseline) and the end of exercise ( $314 \pm 216\%$  of baseline). [ $P_i$ ] increased more rapidly for >CP compared to <CP exercise; however, the increase in [ $P_i$ ] between 3 min ( $449 \pm 141\%$  of baseline) and the end of exercise ( $564 \pm 167\%$  of baseline) was not significant. The [ $P_i$ ] was significantly greater at the end of exercise in >CP compared to <CP ( $P < 0.05$ ). Also, the mean change in [ $P_i$ ] per unit work-rate was also almost twice as great for >CP exercise, but inter-individual variability in the responses precluded the attainment of statistical significance (<CP:  $12.1 \pm 11.9$  vs. >CP:  $21.8 \pm 9.3$   $\% \Delta[P_i] \cdot W^{-1}$ ;  $P > 0.05$ ). The muscle [ $P_i$ ] responses for <CP and >CP exercise are shown in Figure 3 for a representative subject (Panel A) and for the entire group (Panel B).

pH. The two-way ANOVA with repeated measures across intensity and absolute time revealed a significant intensity-by-time interaction ( $F_{5,25} = 6.73$ ;  $P < 0.01$ ) and a significant main effects for time ( $F_{5,25} = 29.40$ ;  $P < 0.01$ ) but not intensity ( $F_{1,5} = 1.69$ ;  $P = 0.26$ ). The subsequent one-way ANOVAs with repeated measures across time indicated the existence of significant differences (<CP,  $F_{5,25} = 9.06$ ;  $P < 0.01$ ; >CP,  $F_{5,25} = 25.07$ ;  $P < 0.01$ ). The ANOVA results were essentially the same when data were expressed as a fraction of the time at the end of exercise. The pertinent statistical comparisons are presented in Table 2 (for absolute time) and Table 3 (for relative time). During <CP exercise, muscle pH increased transiently during the first minute before falling to reach a nadir at  $\sim 3$  min. There was no significant difference in pH from 3 min ( $6.99 \pm 0.05$ ) until the end of exercise ( $7.01 \pm 0.03$ ). Indeed, the slight recovery of pH after  $\sim 6$ -8 min of exercise meant that the

pH at the end of exercise was not significantly different from the resting value. During >CP exercise, however, muscle pH fell precipitously to reach  $6.88 \pm 0.07$  at 6 min, a value which was not significantly different from the value reached at the termination of exercise ( $6.87 \pm 0.10$ ). Muscle pH was significantly lower for >CP compared to <CP exercise at 1, 3, 6 and 10 min and at the end of exercise ( $P < 0.05$ ). The change in pH from rest to end-exercise per unit work-rate was three times greater for >CP exercise (<CP:  $0.003 \pm 0.003$  vs. >CP:  $0.009 \pm 0.004 \Delta\text{pH} \cdot \text{W}^{-1}$ ;  $P < 0.05$ ). The muscle pH responses for <CP and >CP exercise are shown in Figure 4 for a representative subject and for the entire group in Panel A and Panel B, respectively.

## Discussion

The principal novel finding of this investigation was that the muscle metabolic response to exercise (as assessed with  $^{31}\text{P}$ -MRS) differs profoundly according to whether the work-rate is incrementally below or above the CP. The results of the study were, in part, consistent with our hypothesis and support the theory that constant-work-rate exercise above (but not below) the CP results in the utilization of the finite capacity for ATP re-synthesis through non-oxidative pathways until the  $W'$  is exhausted and exhaustion ensues. During exercise that was just below CP, a steady-state in muscle [PCr] and pH was attained within 3 min and a steady state in  $[\text{P}_i]$  was attained within 1 min following the onset of exercise; moreover, the metabolic perturbation at this intensity was relatively slight: after 20 min of exercise, [PCr] had fallen to approximately three-quarters of the baseline value and pH was 7.01 (not significantly different from the value measured at rest). In stark contrast, during exercise that was just above CP, there was a precipitous change in pH and  $[\text{P}_i]$  over the first 3-6 min of exercise, and [PCr] fell progressively with time until subjects were unable to sustain the work-rate. The results therefore indicate that the CP demarcates an exercise intensity domain within which these muscle metabolic variables can be rapidly stabilised and sustained close to resting values from one in which they change precipitously (pH,  $[\text{P}_i]$ ), or progressively ([PCr]), to values that may be limiting to performance (10, 11, 33, 52, 61).

*I: The work-rate vs. time-to-exhaustion relationship and estimation of CP*

That the tolerable duration of high-intensity exercise is hyperbolically related to the work-rate is well established (20, 24, 37, 38). This relationship is very well described by a hyperbolic function both in humans (37, 38, 44) and in other species including the crab (16), the lungless salamander (15), the horse (34) and the mouse (6). It has been posited that the asymptote of the hyperbolic relationship, the CP, reflects the highest oxidative metabolic rate that can be sustained for a long time without fatigue, whereas the curvature constant of the relationship (the  $W'$ ) represents a constant amount of work that can be performed above the CP irrespective of the rate at which the work is done (37, 38, 63). The tolerable duration of >CP exercise is therefore related to the difference between the work-rate being sustained and the CP: the higher the work-rate above the CP, the more rapidly the stored energy sources ( $O_2$  reserves, high-energy phosphates and a source related to anaerobic glycolysis) should be depleted, and the more rapidly the by-products of these reactions (ADP,  $P_i$ ,  $H^+$ , lactate) should accumulate in the contracting muscles and blood (37, 38, 44). While there are numerous assumptions surrounding the CP concept and its predictions, some of which are problematic (21, 39), numerous studies have demonstrated that physiological responses to >CP exercise generally conform to the predictions of the CP concept (14, 19, 23, 29, 35-36, 44-45, 47, 54-55). In particular, it appears that the CP broadly corresponds to the work-rate at the maximal (lactate) steady state (44, 47, 54). The CP has therefore been suggested to demarcate the 'heavy' intensity exercise domain, within which steady-states in  $\dot{V}O_2$  and blood [lactate] are attainable, from the 'severe' intensity exercise domain, within which  $\dot{V}O_2$  and blood [lactate] do not stabilize but continue to rise with time until the  $\dot{V}O_2$  max is reached (27, 42, 44, 65).

In the present study, the work-rate vs. time-to-exhaustion relationship was established from 3-4 constant-work-rate trials performed on different days, with the tolerable duration of exercise ranging from 3-18 min, i.e. consistent with recommendations for accurate assessment of CP (7, 8, 21). Consistent with previous studies (23, 40; *cf.* 17), the non-linear power-time model, the linear work-time model, and the linear 1/time model all produced similar values for CP and  $W'$ . The linear work-rate vs. 1/time model (13, 44, 47), in which the y intercept from the linear regression model gives the CP while the gradient of the relationship gives the  $W'$  (Figure 1), produced  $R^2$  values >0.97 for all subjects (Table 1) and was used to calculate work-rates that were ~ 10% below and ~ 10% above the CP.

## II: <sup>31</sup>P-MRS assessment of muscle metabolic responses to <CP and >CP exercise

To our knowledge, the present study is the first to investigate the muscle metabolic responses to constant-work-rate exercise performed below and above the CP. During exercise performed at a work-rate that was ~ 10% below the CP (~ 2 W below CP on average), it is clear that a steady-state in muscle metabolites is achieved relatively rapidly (i.e. *within* ~ 3 min) and that the muscle metabolic perturbation is relatively slight (Figures 2-4). For example, at the end of exercise, muscle [PCr] had fallen, on average, to 68% of the baseline value and pH had fallen by 0.05 units from the resting value. In all cases, the subjects were able to complete 20 min of <CP exercise without significant duress. These data support the suggestion that exercise <CP can be sustained for an appreciable duration without significant fatigue development (37, 38).

The muscle metabolic response to constant-work-rate exercise performed ~ 10% (~ 2 W) above the CP was strikingly different to the response observed below the CP (Figures 2-4). During exercise >CP, [PCr] fell progressively throughout exercise, reaching ~ 26% of the baseline value at the end of exercise (Figure 2); this continued fall in [PCr] is consistent with the predictions of the CP concept (37, 38, 44). Moreover, pH fell abruptly (reaching a value of 6.87 at exhaustion) and [P<sub>i</sub>] increased rapidly following the onset of exercise (Figures 3-4). The non-linearity of the muscle metabolic responses to exercise performed below and above the CP is underlined when the change in the metabolic variables from rest to end-exercise is expressed relative to the change in work-rate: that is, for the same unit change in work-rate, the change in pH was three times as great and the change in [PCr] and [P<sub>i</sub>] was almost twice as great, when the exercise was >CP.

The time-to-exhaustion during the >CP exercise bout averaged 14.7 min, ranging from a minimum of 6.9 min to a maximum of 23.8 min. Clearly, exercise tolerance is limited during >CP exercise. Clues to the mechanism(s) responsible for the more rapid muscular fatigue development during >CP exercise can be obtained from the present study. Classically, fatigue during >CP exercise has been attributed to a depletion of the W', which is considered to be a finite energy store that is predominantly (i.e., energy contained in the high-energy phosphates and a source linked to anaerobic glycolysis), although not exclusively (i.e., energy equivalent contained in stored O<sub>2</sub> reserves in myoglobin and venous blood), 'anaerobic' (37, 38). Data from the present study certainly indicate a

significant depletion of the high-energy phosphates: [PCr] fell by an average of 74% at exhaustion during >CP exercise. Several micro-dissection studies have demonstrated that the depletion of muscle high-energy phosphates during exercise is heterogeneous, with the heterogeneity appearing to be related to muscle fibre type (32, 52-53). These studies indicate that a reduction in the ability to generate muscle power during exercise might ultimately be attributed to fatigue and reduced force production in a relatively small population of muscle fibres (32, 53). Therefore, it is possible that [PCr] (and perhaps [ATP] also) fell sufficiently low in at least some of the recruited muscle fibres during exercise >CP that the required muscle force could not be maintained. It should also be considered that during >CP exercise, 'higher-order' muscle fibres are progressively recruited to maintain muscle power output as the initially recruited fibres become fatigued until, eventually, the maximal voluntary contraction force falls below the force required for the work-rate (5, 59).

The limited exercise tolerance >CP might also be attributed to the extent, or the rate, of accumulation of metabolites that have been associated with the fatigue process, such as  $P_i$  and  $H^+$ . Historically, a reduction in muscle pH has been linked to the fatigue process through  $H^+$  ion competition with  $Ca^{2+}$  for binding to troponin, interference with  $Ca^{2+}$  release from the sarcoplasmic reticulum, and inhibition of phosphofructokinase (11, 12, 57). More recently, the role of acidosis in muscle fatigue has been questioned while the possible role of  $P_i$  accumulation (specifically in its diprotonated form) has received increasing attention (1, 10, 41, 48, 61, 62, 66). In the present study, muscle  $[P_i]$  and pH changed precipitously following the onset of >CP exercise to values which might potentially have limited muscle contractile function. However,  $[P_i]$  did not increase significantly between 3 minutes and the end of exercise, and pH did not fall significantly between 6 minutes and the end of exercise. The depletion of muscle [PCr] would therefore appear to have a stronger temporal relationship with muscle fatigue development during exercise >CP. It cannot be ruled out, however, that  $[P_i]$  and  $[H^+]$  continued to increase with time to the point of exhaustion in individual muscle fibres and that this was 'masked' by inter-fibre, inter-subject and measurement variability, and it remains possible that metabolite accumulation was responsible for, or at least contributed to, the limited exercise tolerance >CP. In this respect, it is noteworthy that the change in [PCr] and  $[P_i]$ , which would be expected to be stoichiometric, demonstrated somewhat different temporal profiles during >CP exercise: that is, the continued fall of [PCr] with time during >CP exercise was not mirrored by a continued rise in  $[P_i]$ ; rather,  $[P_i]$  rose rapidly following the onset of exercise to reach a high but stable value. It is possible that this was caused by difficulty in precisely quantifying the



changes in  $[P_i]$  during high-intensity exercise in which a low pH can result in a broadening or splitting of the  $[P_i]$  peak or be due to changes in the longitudinal relaxation time ( $T_1$ ) for  $P_i$  (49, 50).

Muscle fatigue during high-intensity exercise is highly complex and is likely explained by a combination of interacting mechanisms (2, 10, 33, 41, 43, 48, 61). However, based on the muscle metabolic responses to <CP and >CP exercise reported in the present study, it seems reasonable to suggest that the CP might differentiate exercise intensities that are principally limited by high-energy phosphate depletion and/or the effects of metabolite accumulation on cross-bridge function (>CP) from those that are chiefly limited by the availability of oxidative substrate (chiefly muscle glycogen) and/or hyperthermia or perhaps central fatigue (<CP), as was suggested in some of the original CP investigations (38, 44).

The muscle [PCr] response to exercise has been reported to have similar kinetics to that of pulmonary  $\dot{V}O_2$  and this has been suggested to reflect feedback control of oxidative phosphorylation through one or more of the reactants and/or products of high-energy phosphate hydrolysis (3, 49, 50). It is therefore of interest to note that the response profiles of muscle [PCr] in the present study had similar characteristics to what would be expected for pulmonary  $\dot{V}O_2$  during <CP and >CP exercise (42, 65). The present data also confirm the existence of a [PCr] 'slow component' during high-intensity constant-work-rate exercise (18, 30, 50, 51): the change in [PCr] between 3 and 6 minutes of exercise was significantly greater for >CP compared to <CP exercise. As recognized previously, the existence of a [PCr] slow component implies an increasing phosphate cost of force production as exercise proceeds, an effect that might be attributed to the effects of fatigue (on the ATP cost of ion pumping, for example) and/or an increasing contribution of type II muscle fibres to force production (18, 46, 50). Conley *et al.* (9) have argued that the highest sustainable oxidative metabolic rate (i.e. ostensibly the CP) is determined by the interaction between glycolytic flux and oxidative phosphorylation and can be understood with reference to the creatine kinase reaction. Specifically, they propose that the increased  $[H^+]$  resulting from a greater activation of glycolysis during high-intensity exercise limits the extent to which [ADP] (believed to be a key signal activating mitochondrial respiration) can rise such that the highest steady-state  $\dot{V}O_2$  (and the highest sustainable work-rate) will occur at approximately 80% of the respective maxima achieved during an incremental ( $\dot{V}O_{2\max}$ ) test (see ref. 4). However, [ADP] can be made to increase, allowing exercise to be continued for a short period (~ 5-10 min) and  $\dot{V}O_2$  to reach its maximal rate, if [PCr] is reduced

to offset the fall in pH (9, 31). The continued fall in muscle [PCr] during exercise >CP might therefore be necessary to enable the appropriate [ADP] stimulus to oxidative phosphorylation but is also characteristic of a fatiguing muscle with a limited capacity for continued force generation.

### *Perspectives and Significance*

We have demonstrated that the dynamics of the muscle metabolic response to exercise (as assessed using  $^{31}\text{P}$ -MRS) can be differentiated according to whether the exercise is performed just below or just above the CP, therefore providing empirical verification of the CP construct. Specifically, the CP appears to demarcate a range of work-rates within which muscle [PCr],  $[\text{P}_i]$  and pH can be rapidly stabilised and sustained close to resting values, from those within which they change precipitously (pH,  $[\text{P}_i]$ ), or progressively ([PCr]), to values that might predispose the muscle to the development of fatigue. The CP concept therefore has theoretical and practical utility in exercise physiology as a model for exploring the mechanistic bases to muscular fatigue and the determinants of human exercise tolerance.

## Acknowledgements

The authors thank Ms. Anni Vanhatalo for assistance with non-linear modeling of the power-time relationship.

## References

1. **Allen DG and Westerblad H.** Role of phosphate and calcium stores in muscle fatigue. *J Physiol* 536: 657-665, 2001.
2. **Bangsbo J, Madsen K, Kiens B, and Richter EA.** Effect of muscle acidity on muscle metabolism and fatigue during intense exercise in man. *J Physiol* 495: 587-596, 1996.
3. **Barstow TJ, Buchthal S, Zanconato S, and Cooper DM.** Muscle energetics and pulmonary oxygen uptake kinetics during moderate exercise. *J Appl Physiol* 77: 1742-1749, 1994.
4. **Barstow TJ, Buchthal SD, Zanconato S, and Cooper DM.** Changes in potential controllers of human skeletal muscle respiration during incremental calf exercise. *J Appl Physiol* 77: 2169-2176, 1994.
5. **Bigland-Ritchie B, Cafarelli E, and Vøllestad NK.** Fatigue of submaximal static contractions. *Acta Physiol Scand Suppl* 556:137-148, 1986.
6. **Billat VL, Mouisel E, Roblot N, and Melki J.** Inter- and intra-strain variation in mouse critical running speed. *J Appl Physiol* 98: 1258-1263, 2005.
7. **Bishop D and Jenkins DG.** The influence of recovery duration between periods of exercise on the critical power function. *Eur J Appl Physiol Occup Physiol* 72: 115-120, 1995.
8. **Bishop D, Jenkins DG and Howard A.** The critical power function is dependent on the duration of the predictive exercise tests chosen. *Int J Sports Med* 19: 125-129, 1998.
9. **Conley KE, Kemper WF, and Crowther GJ.** Limits to sustainable muscle performance: interaction between glycolysis and oxidative phosphorylation. *J Exp Biol* 204: 3189-3194, 2001.
10. **Cooke R, Franks K, Luciani GB, and Pate E.** The inhibition of rabbit skeletal muscle contraction by hydrogen ions and phosphate. *J Physiol* 395: 77-97, 1988.
11. **Fitts RH.** Cellular mechanisms of muscle fatigue. *Physiol Rev* 74: 49-94, 1994.

12. **Fuchs F, Reddy Y, and Briggs FN.** The interaction of cations with the calcium-binding site of troponin. *Biochim Biophys Acta* 221: 407-409, 1970.
13. **Fukuba Y, Miura A, Endo M, Kan A, Yanagawa K, and Whipp BJ.** The curvature constant parameter of the power-duration curve for varied-power exercise. *Med Sci Sports Exerc* 35: 1413-1418, 2003.
14. **Fukuba Y and Whipp BJ.** A metabolic limit on the ability to make up for lost time in endurance events. *J Appl Physiol* 87: 853-861, 1999.
15. **Full RJ.** Locomotion without lungs: energetics and performance of a lungless salamander. *Am J Physiol* 251: R775-780, 1986.
16. **Full RJ and Herreid CF.** Aerobic response to exercise of the fastest land crab. *Am J Physiol* 244: R530-536, 1983.
17. **Gaesser GA, Carnevale TJ, Garfinkel A, Walter DO, and Womack CJ.** Estimation of critical power with nonlinear and linear models. *Med Sci Sports Exerc* 27: 1430-1438, 1995.
18. **Haseler LJ, Kindig CA, Richardson RS, and Hogan MC.** The role of oxygen in determining phosphocreatine onset kinetics in exercising humans. *J Physiol* 558: 985-992, 2004.
19. **Heubert RA, Billat VL, Chassaing P, Bocquet V, Morton RH, Koralsztejn JP, and di Prampero PE.** Effect of a previous sprint on the parameters of the work-time to exhaustion relationship in high intensity cycling. *Int J Sports Med* 26: 583-592, 2005.
20. **Hill AV.** Muscular Movement in Man. McGraw-Hill, New York, 1927.
21. **Hill DW.** The critical power concept. A review. *Sports Med* 16: 237-254, 1993.
22. **Hill DW, Alain C, and Kennedy MD.** Modeling the relationship between velocity and time to fatigue in rowing. *Med Sci Sports Exerc* 35: 2098-2105, 2003.
23. **Hill DW.** The relationship between power and time to fatigue in cycle ergometer exercise. *Int J Sports Med* 25: 357-361, 2004.

24. **Hughson RL, Orok CJ, and Staudt LE.** A high velocity treadmill running test to assess endurance running potential. *Int J Sports Med* 5: 23-25, 1984.
  
25. **Jenkins DG and Quigley BM.** Endurance training enhances critical power. *Med Sci Sports Exerc* 24: 1283-1289, 1992.
  
26. **Jenkins DG and Quigley BM.** The influence of high-intensity exercise training on the Wlim-Tlim relationship. *Med Sci Sports Exerc* 25: 275-282, 1993.
  
27. **Jones AM and Doust JH.** Limitations to submaximal exercise performance. In *Kinanthropometry and Exercise Physiology Laboratory Manual: Tests, Procedures and Data*, vol 2, *Exercise Physiology* (2<sup>nd</sup> Edition), ed Eston RG and Reilly TP, pp. 235-262, Routledge, London, 2001.
  
28. **Jones AM and Poole DC.** Introduction to oxygen uptake kinetics and historical development of the discipline. In *Oxygen Uptake Kinetics in Sport, Exercise and Medicine*, ed Jones AM and Poole DC, pp. 3-35, Routledge, London and New York, 2005.
  
29. **Jones AM and Whipp BJ.** Bioenergetic constraints on tactical decision making in middle distance running. *Br J Sports Med* 36: 102-104, 2002.
  
30. **Jones AM, Wilkerson DP, Berger NJ, and Fulford J.** Influence of endurance training on muscle [PCr] kinetics during high-intensity exercise. *Am J Physiol Regul Integr Comp Physiol* 293: R392-401, 2007.
  
31. **Jubrias SA, Crowther GJ, Shankland EG, Gronka RK, and Conley KE.** Acidosis inhibits oxidative phosphorylation in contracting human skeletal muscle in vivo. *J Physiol* 553: 589-599, 2003.
  
32. **Karatzafieri C, de Haan A, van Mechelen W, and Sargeant AJ.** Metabolism changes in single human fibres during brief maximal exercise. *Exp Physiol* 86: 411-415, 2001.
  
33. **Katz A, Sahlin H, and Henriksson J.** Muscle ATP turnover rate during isometric contraction in humans. *J Appl Physiol* 60: 1839-1842, 1986.

34. **Lauderdale MA and Hinchcliff KW.** Hyperbolic relationship between time-to-fatigue and workload. *Equine Vet J Suppl* 30: 586-590, 1999.
35. **Miura A, Kino F, Kajitani S, Sato H, and Fukuba Y.** The effect of oral creatine supplementation on the curvature constant parameter of the power-duration curve for cycle ergometry in humans. *Jpn J Physiol* 49: 169-174, 1999.
36. **Miura A, Sato H, Sato H, Fukuba Y, and Whipp BJ.** The effect of glycogen depletion on the curvature constant parameter of the power-duration curve for cycle ergometry. *Ergonomics* 43: 133-141, 2000.
37. **Monod H and Scherrer J.** The work capacity of a synergic muscle group. *Ergonomics* 8: 329-338, 1965.
38. **Moritani T, Nagata A, deVries HA, and Muro M.** Critical power as a measure of physical work capacity and anaerobic threshold. *Ergonomics* 24: 339-350, 1981.
39. **Morton RH.** The critical power and related whole-body bioenergetic models. *Eur J Appl Physiol* 96: 339-354, 2006.
40. **Morton RH, Green S, Bishop D, and Jenkins DG.** Ramp and constant power trials produce equivalent critical power estimates. *Med Sci Sports Exerc* 29: 833-836, 1997.
41. **Nosek TM, Fender KY, and Godt RE.** It is diprotonated inorganic phosphate that depresses force in skinned skeletal muscle fibers. *Science* 236: 191-193, 1987.
42. **Özyener F, Rossiter HB, Ward SA, and Whipp BJ.** Influence of exercise intensity on the on- and off-transient kinetics of pulmonary oxygen uptake in humans. *J Physiol* 533: 891-902, 2001.
43. **Pedersen TH, Nielsen OB, Lamb GD, and Stephenson DG.** Intracellular acidosis enhances the excitability of working muscle. *Science* 305: 1144-1147, 2004.
44. **Poole DC, Ward SA, Gardner GW, and Whipp BJ.** Metabolic and respiratory profile of the upper limit for prolonged exercise in man. *Ergonomics* 31: 1265-1279, 1988.

45. **Poole DC, Ward SA, and Whipp BJ.** The effects of training on the metabolic and respiratory profile of high-intensity cycle ergometer exercise. *Eur J Appl Physiol Occup Physiol* 59: 421-429, 1990.
46. **Pringle JS, Doust JH, Carter H, Tolfrey K, Campbell IT, Sakkas GK, and Jones AM.** Oxygen uptake kinetics during moderate, heavy and severe intensity "submaximal" exercise in humans: the influence of muscle fibre type and capillarisation. *Eur J Appl Physiol* 89: 289-300, 2003.
47. **Pringle JS and Jones AM.** Maximal lactate steady state, critical power and EMG during cycling. *Eur J Appl Physiol* 88: 214-226, 2002.
48. **Robergs RA, Ghiasvand F and Parker D.** Biochemistry of exercise-induced metabolic acidosis. *Am J Physiol Regul Integr Comp Physiol* 287: R502-516, 2004.
49. **Rossiter HB, Howe FA, and Ward SA.** Intramuscular phosphate and pulmonary  $\dot{V}_{O_2}$  kinetics during exercise: implications for control of skeletal muscle oxygen consumption. In *Oxygen Uptake Kinetics in Sport, Exercise and Medicine*, ed Jones AM and Poole DC, pp. 154-184, Routledge, London and New York, 2005.
50. **Rossiter HB, Ward SA, Howe FA, Kowalchuk JM, Griffiths JR, and Whipp BJ.** Dynamics of intramuscular  $^{31}\text{P}$ -MRS  $\text{P}_i$  peak splitting and the slow components of PCr and  $\text{O}_2$  uptake during exercise. *J Appl Physiol* 93: 2059-2069, 2002.
51. **Rossiter HB, Ward SA, Kowalchuk JM, Howe FA, Griffiths JR, and Whipp BJ.** Effects of prior exercise on oxygen uptake and phosphocreatine kinetics during high-intensity knee-extension exercise in humans. *J Physiol* 537: 291-303, 2001.
52. **Sahlin K, Soderlund K, Tonkonogi M, and Hirakoba K.** Phosphocreatine content in single fibers of human muscle after sustained submaximal exercise. *Am J Physiol* 273: C172-178, 1997.
53. **Sant'Ana Pereira JA, Sargeant AJ, Rademaker AC, de Haan A, and van Mechelen W.** Myosin heavy chain isoform expression and high energy phosphate content in human muscle fibres at rest and post-exercise. *J Physiol* 496: 583-588, 1996.
54. **Smith CG and Jones AM.** The relationship between critical velocity, maximal lactate steady-state velocity and lactate turnpoint velocity in runners. *Eur J Appl Physiol* 85: 19-26, 2001.



55. **Smith JC, Stephens DP, Hall EL, Jackson AW, and Earnest CP.** Effect of oral creatine ingestion on parameters of the work rate-time relationship and time to exhaustion in high-intensity cycling. *Eur J Appl Physiol Occup Physiol* 77: 360-365, 1998.
56. **Taylor DJ, Bore PJ, Styles P, Gadian DG, and Radda GK.** Bioenergetics of intact human muscle. A <sup>31</sup>P nuclear magnetic resonance study. *Mol Biol Med* 1: 77-94, 1983.
57. **Trivedi B and Danforth WH.** Effect of pH on the kinetics of frog muscle phosphofructokinase. *J Biol Chem* 241: 4110-4112, 1966.
58. **Vanhamme L, van den Boogaart A, and Van Huffel S.** Improved method for accurate and efficient quantification of MRS data with use of prior knowledge. *J Magn Reson* 129: 35-43, 1997.
59. **Vøllestad NK, Vaage O, and Hermansen L.** Muscle glycogen depletion patterns in type I and subgroups of type II fibres during prolonged severe exercise in man. *Acta Physiol Scand* 122: 433-441, 1984.
60. **Wakayoshi K, Ikuta K, Yoshida T, Udo M, Moritani T, Mutoh Y, and Miyashita M.** Determination and validity of critical velocity as an index of swimming performance in the competitive swimmer. *Eur J Appl Physiol Occup Physiol* 64: 153-157, 1992.
61. **Westerblad H and Allen DG.** Cellular mechanisms of skeletal muscle fatigue. *Adv Exp Med Biol* 538: 563-570, 2003.
62. **Westerblad H, Allen DG, and Lannergren J.** Muscle fatigue: lactic acid or inorganic phosphate the major cause? *News Physiol Sci* 17: 17-21, 2002.
63. **Whipp BJ, Huntsman DJ, Stoner N, Lamarra N, and Wasserman K.** A constant which determines the duration of tolerance to high-intensity work. *Fed Proc* 41: 1591, 1981.
64. **Whipp BJ and Rossiter HB.** The kinetics of oxygen uptake: physiological inferences from the parameters. In *Oxygen Uptake Kinetics in Sport, Exercise and Medicine*, ed Jones AM and Poole DC, pp. 62-94, Routledge, London and New York, 2005.

65. **Wilkerson DP, Koppo K, Barstow TJ, and Jones AM.** Effect of work rate on the functional 'gain' of Phase II pulmonary O<sub>2</sub> uptake response to exercise. *Respir Physiol Neurobiol* 142: 211-223, 2004.
66. **Wilson JR, McCully KK, Mancini DM, Boden B, and Chance B.** Relationship of muscular fatigue to pH and diprotonated P<sub>i</sub> in humans: a <sup>31</sup>P-NMR study. *J Appl Physiol* 64: 2333-2339, 1988.

**Table 1: Estimates of the subjects' CP and W' and work rates applied in the <CP and >CP conditions**

| <b>Subject</b>   | <b>CP (W)</b> | <b>W' (kJ)</b>     | <b>R<sup>2</sup></b> | <b>10%&lt;CP (W)</b> | <b>10%&gt;CP (W)</b> |
|------------------|---------------|--------------------|----------------------|----------------------|----------------------|
| #1               | 20            | 1.71               | 1.00                 | 18                   | 22                   |
| #2               | 19            | 1.90               | 0.97                 | 17                   | 21                   |
| #3               | 21            | 1.75               | 1.00                 | 19                   | 22                   |
| #4               | 21            | 2.08               | 1.00                 | 18                   | 23                   |
| #5               | 17            | 3.01               | 0.98                 | 15                   | 19                   |
| #6               | 20            | 1.23               | 0.99                 | 18                   | 22                   |
| <b>Mean ± SD</b> | <b>20 ± 2</b> | <b>1.95 ± 0.59</b> | <b>0.99 ± 0.01</b>   | <b>18 ± 1</b>        | <b>22 ± 1</b>        |

Data presented are derived from the linear work-rate vs. 1/time relationship (Model 3; see text).

**Table 2: Mean  $\pm$  SD  $^{31}\text{P}$ -MRS derived muscle metabolic responses to exercise performed below and above critical power**

|                         | <b>0 min</b>    | <b>1 min</b>               | <b>3 min</b>                   | <b>6 min</b>                       | <b>10 min</b>                    | <b>End-Exercise</b>                |
|-------------------------|-----------------|----------------------------|--------------------------------|------------------------------------|----------------------------------|------------------------------------|
| <b>[PCr], &lt;CP, %</b> | 100 $\pm$ 0     | 82 $\pm$ 11 <sup>1</sup>   | 75 $\pm$ 13 <sup>1,2</sup>     | 71 $\pm$ 12 <sup>1,2</sup>         | 71 $\pm$ 11 <sup>1,2</sup>       | 68 $\pm$ 11 <sup>1,2</sup>         |
| <b>[PCr], &gt;CP, %</b> | 100 $\pm$ 0     | 67 $\pm$ 16 <sup>*,1</sup> | 48 $\pm$ 19 <sup>*,1,2</sup>   | 34 $\pm$ 20 <sup>*,1,2,3</sup>     | 30 $\pm$ 21 <sup>*,1,2,3,4</sup> | 26 $\pm$ 16 <sup>*,1,2,3,4,5</sup> |
| <b>[Pi], &lt;CP, %</b>  | 100 $\pm$ 0     | 270 $\pm$ 56 <sup>1</sup>  | 341 $\pm$ 104 <sup>1</sup>     | 355 $\pm$ 145 <sup>1</sup>         | 322 $\pm$ 152 <sup>1</sup>       | 314 $\pm$ 216                      |
| <b>[Pi], &gt;CP, %</b>  | 100 $\pm$ 0     | 330 $\pm$ 78 <sup>1</sup>  | 449 $\pm$ 141 <sup>1,2</sup>   | 538 $\pm$ 170 <sup>1,2</sup>       | 529 $\pm$ 214 <sup>1,2</sup>     | 564 $\pm$ 167 <sup>*,1,2</sup>     |
| <b>pH, &lt;CP</b>       | 7.06 $\pm$ 0.04 | 7.08 $\pm$ 0.01            | 6.99 $\pm$ 0.05 <sup>2</sup>   | 6.98 $\pm$ 0.05 <sup>2</sup>       | 6.99 $\pm$ 0.04 <sup>2</sup>     | 7.01 $\pm$ 0.03 <sup>2,4,5</sup>   |
| <b>pH, &gt;CP</b>       | 7.07 $\pm$ 0.03 | 7.05 $\pm$ 0.04            | 6.92 $\pm$ 0.07 <sup>1,2</sup> | 6.88 $\pm$ 0.07 <sup>*,1,2,3</sup> | 6.88 $\pm$ 0.10 <sup>1,2,3</sup> | 6.87 $\pm$ 0.10 <sup>*,1,2</sup>   |

\* = significantly different from <CP exercise for the corresponding time point ( $P<0.05$ ); <sup>1</sup> = significantly different from 0 min ( $P<0.05$ ); <sup>2</sup> = significantly different from 1 min ( $P<0.05$ ); <sup>3</sup> = significantly different from 3 min ( $P<0.05$ ); <sup>4</sup> = significantly different from 6 min ( $P<0.05$ ); and, <sup>5</sup> = significantly different from 10 min ( $P<0.05$ ) over time within a condition (<CP or >CP).

**Table 3: Mean  $\pm$  SD  $^{31}\text{P}$ -MRS derived muscle metabolic responses to exercise performed below and above critical power when data are normalized for differences in the time to end-exercise**

|                         | <b>0%</b>                    | <b>25%</b>                   | <b>50%</b>                     | <b>75%</b>                     | <b>100%</b>                      |
|-------------------------|------------------------------|------------------------------|--------------------------------|--------------------------------|----------------------------------|
| <b>[PCr], &lt;CP, %</b> | 100 $\pm$ 0                  | 73 $\pm$ 14 <sup>1</sup>     | 71 $\pm$ 11 <sup>1</sup>       | 67 $\pm$ 11 <sup>1</sup>       | 68 $\pm$ 11 <sup>1</sup>         |
| <b>[PCr], &gt;CP, %</b> | 100 $\pm$ 0                  | 49 $\pm$ 23 <sup>*,1</sup>   | 37 $\pm$ 20 <sup>*,1,2</sup>   | 32 $\pm$ 16 <sup>*,1,2</sup>   | 26 $\pm$ 16 <sup>*,1,2,3,4</sup> |
| <b>[Pi], &lt;CP, %</b>  | 100 $\pm$ 0                  | 360 $\pm$ 130 <sup>1</sup>   | 322 $\pm$ 152 <sup>1</sup>     | 308 $\pm$ 159 <sup>1</sup>     | 314 $\pm$ 216                    |
| <b>[Pi], &gt;CP, %</b>  | 100 $\pm$ 0                  | 399 $\pm$ 125 <sup>1</sup>   | 473 $\pm$ 171 <sup>1</sup>     | 496 $\pm$ 200 <sup>1</sup>     | 564 $\pm$ 167 <sup>*,1</sup>     |
| <b>pH, &lt;CP</b>       | 7.06 $\pm$ 0.04              | 6.99 $\pm$ 0.05              | 6.99 $\pm$ 0.04                | 7.01 $\pm$ 0.03                | 7.01 $\pm$ 0.03 <sup>3</sup>     |
| <b>pH, &gt;CP</b>       | 7.07 $\pm$ 0.03 <sup>1</sup> | 6.93 $\pm$ 0.07 <sup>1</sup> | 6.88 $\pm$ 0.08 <sup>*,1</sup> | 6.89 $\pm$ 0.08 <sup>*,1</sup> | 6.87 $\pm$ 0.10 <sup>*,1</sup>   |

\* = significantly different from <CP exercise for the corresponding time point ( $P < 0.05$ ); <sup>1</sup> = significantly different from 0% ( $P < 0.05$ ); <sup>2</sup> = significantly different from 25% ( $P < 0.05$ ); <sup>3</sup> = significantly different from 50% ( $P < 0.05$ ); and, <sup>4</sup> = significantly different from 75% ( $P < 0.05$ ), over time within a condition (<CP or >CP).

## Figure Legends

Figure 1: Example of the estimation of CP in a representative subject. Panel A illustrates the hyperbolic nature of the relationship between work-rate and the time-to-exhaustion at that work-rate; Panel B shows the linear transformation of the relationship when work-rate is plotted against the inverse of the time-to-exhaustion. In this formulation, the slope of the regression line represents the finite amount of work that can be performed above CP (i.e. the  $W'$ ) while the y axis intercept gives the CP.

Figure 2: Muscle [PCr] responses to <CP (closed symbols) and >CP (open symbols) knee-extension exercise. Panel A shows the muscle [PCr] responses of a representative subject (#1) during <CP exercise (18 W) and >CP exercise (22 W; time-to-exhaustion of 11.5 minutes). Panel B shows the mean  $\pm$  SEM muscle [PCr] responses for the whole group with the data expressed as a percentage of the time to end-exercise (20 min for <CP and exhaustion for >CP). Muscle [PCr] was significantly lower for >CP exercise at 25%, 50%, 75% and 100% of the time to end-exercise ( $* = P < 0.01$ ). For <CP exercise, [PCr] fell significantly from 0% to 25% of the time to end-exercise ( $P < 0.01$ ), but did not change significantly thereafter. For >CP exercise, [PCr] fell significantly at each consecutive time point ( $P < 0.05$ ), except between 50% and 75% of the time to end-exercise ( $P = 0.12$ ). See text for further information.

Figure 3: Muscle [ $P_i$ ] responses to <CP (closed symbols) and >CP (open symbols) knee-extension exercise. Panel A shows the muscle [ $P_i$ ] responses of a representative subject (#1) during <CP and >CP exercise. Panel B shows the mean  $\pm$  SEM muscle [ $P_i$ ] responses for the whole group. Muscle [ $P_i$ ] was significantly higher for >CP exercise only at the end of exercise ( $* = P < 0.05$ ). Muscle [ $P_i$ ] increased significantly from 0% to 25% of the time to end-exercise ( $P < 0.01$ ) but did not change significantly thereafter at either work-rate. See text for further information.

Figure 4: Muscle pH responses to <CP (closed symbols) and >CP (open symbols) knee-extension exercise. Panel A shows the muscle pH responses of a representative subject (#1) during <CP and >CP exercise. Panel B shows the mean  $\pm$  SEM muscle pH responses for the whole group. Muscle pH was significantly lower for >CP exercise at 25%, 50%, 75% and 100% of the time to end-exercise ( $* = P < 0.05$ ). For <CP exercise, pH was not significantly different between 0% and 100%

of the time to end-exercise, except for a significant increase in pH between 50% and 100% of the time to end-exercise ( $P<0.05$ ). For >CP exercise, pH fell significantly from 0% to 25% of the time to end-exercise ( $P<0.01$ ) but did not change significantly with time thereafter. See text for further information.

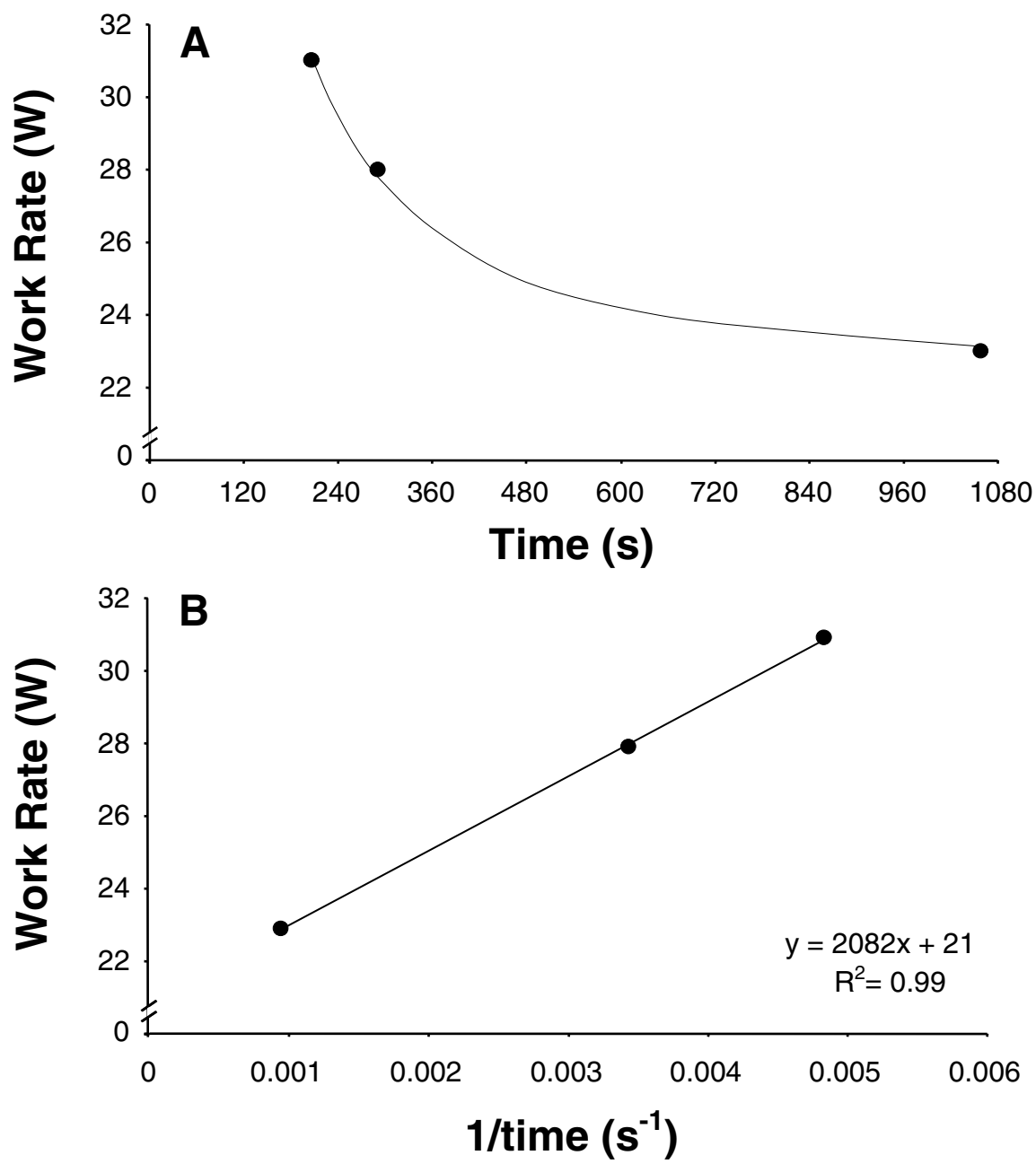


Figure 1



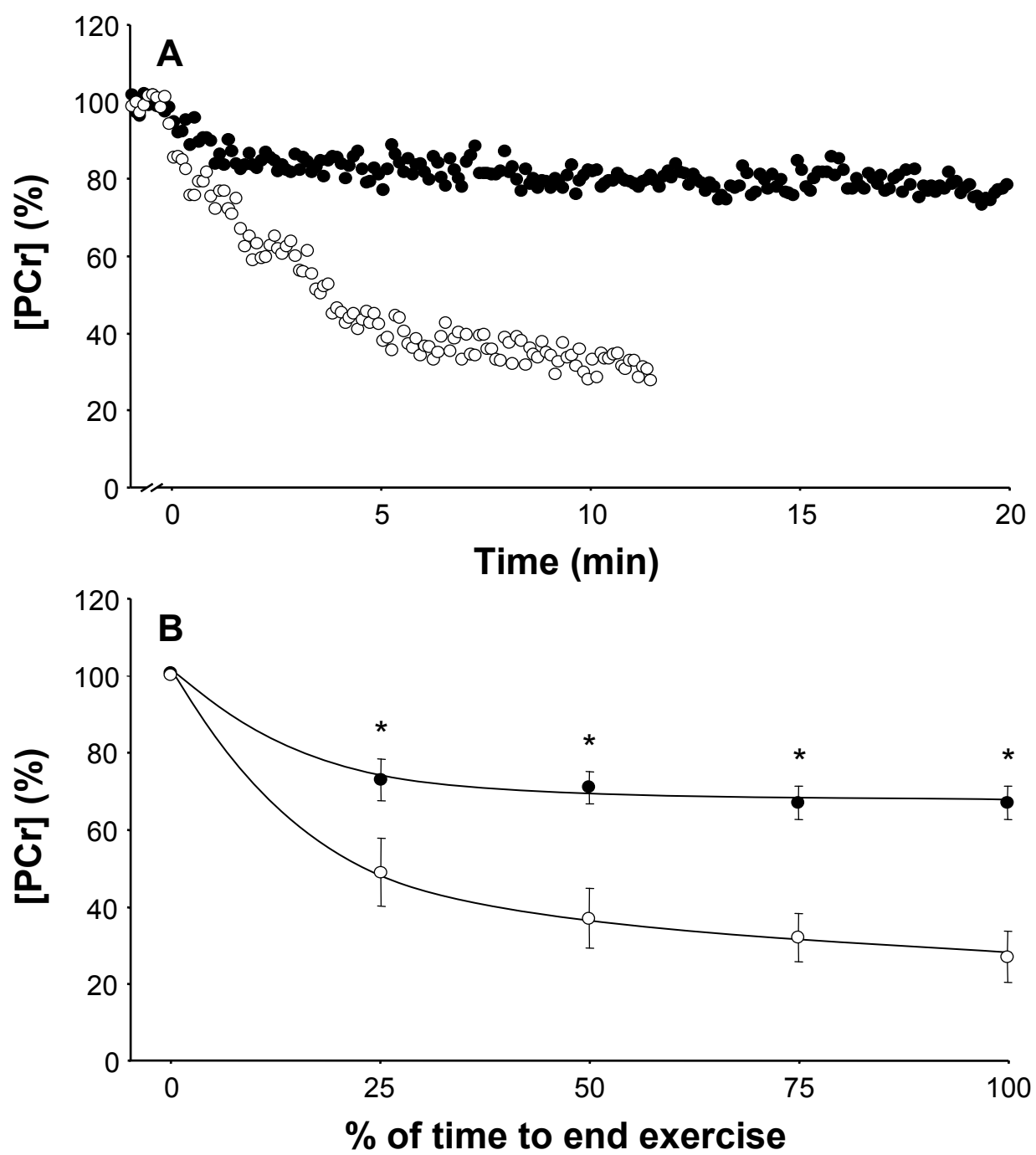


Figure 2

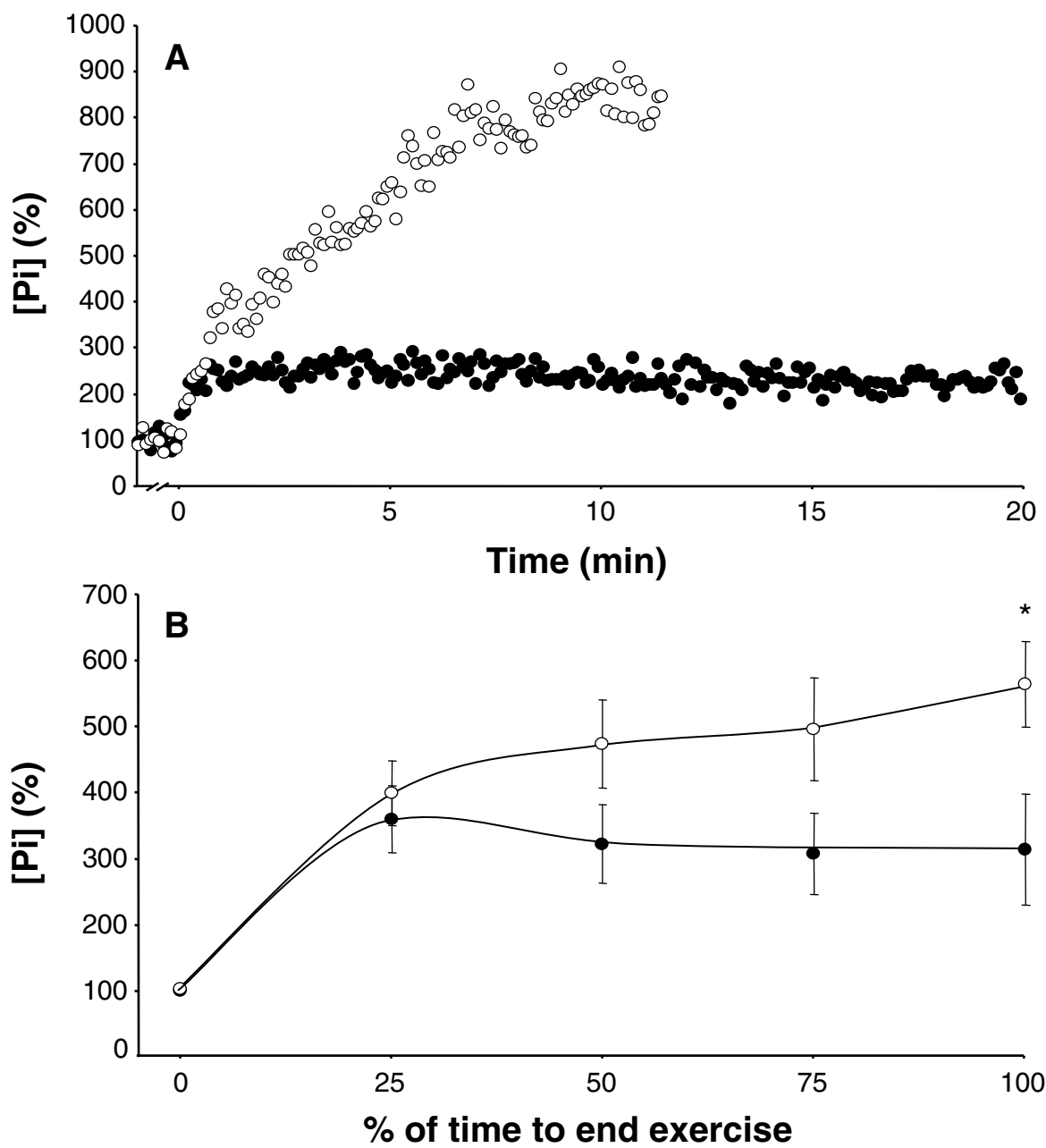


Figure 3

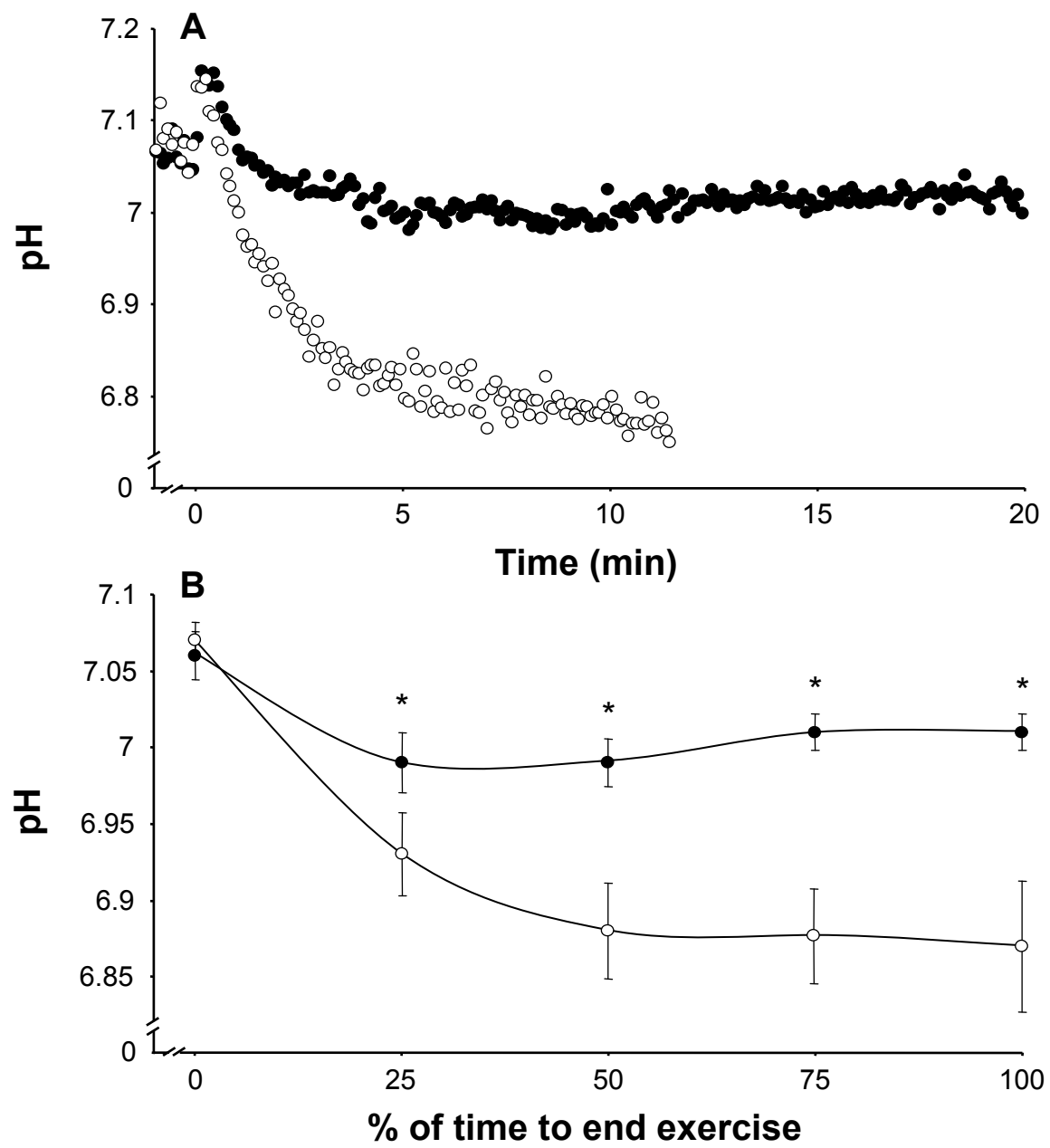


Figure 4